Barrel & Brew
Digital Transformation

Business Requirements & Gap Analysis — June 2026

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Executive Summary

The challenge at a glance

Problem: Barrel & Brew's rapid growth has exposed critical weaknesses across production, quality, forecasting, and communication. Manual processes, outdated equipment, and disconnected systems are preventing the business from scaling sustainably.

30%

Overtime increase due to manual workload

2

Full batches discarded from quality failures

0

Real-time dashboards or data visibility tools

Proposed Solution: An integrated Brewery Management System (BMS) with automated forecasting, digital quality control, IoT-enabled equipment, and real-time cross-team dashboards.

Gap Analysis: Current State vs Desired State

6 capability gaps identified

1 Reliable Production

No automated scheduling or real-time production visibility

2 High Quality

No digital QC — reliance on manual taste tests only

3 Empowered Staff

Staff overwhelmed; knowledge locked in notebooks & memory

4 Streamlined Ops

Silos between Sales & Production; no workflow automation

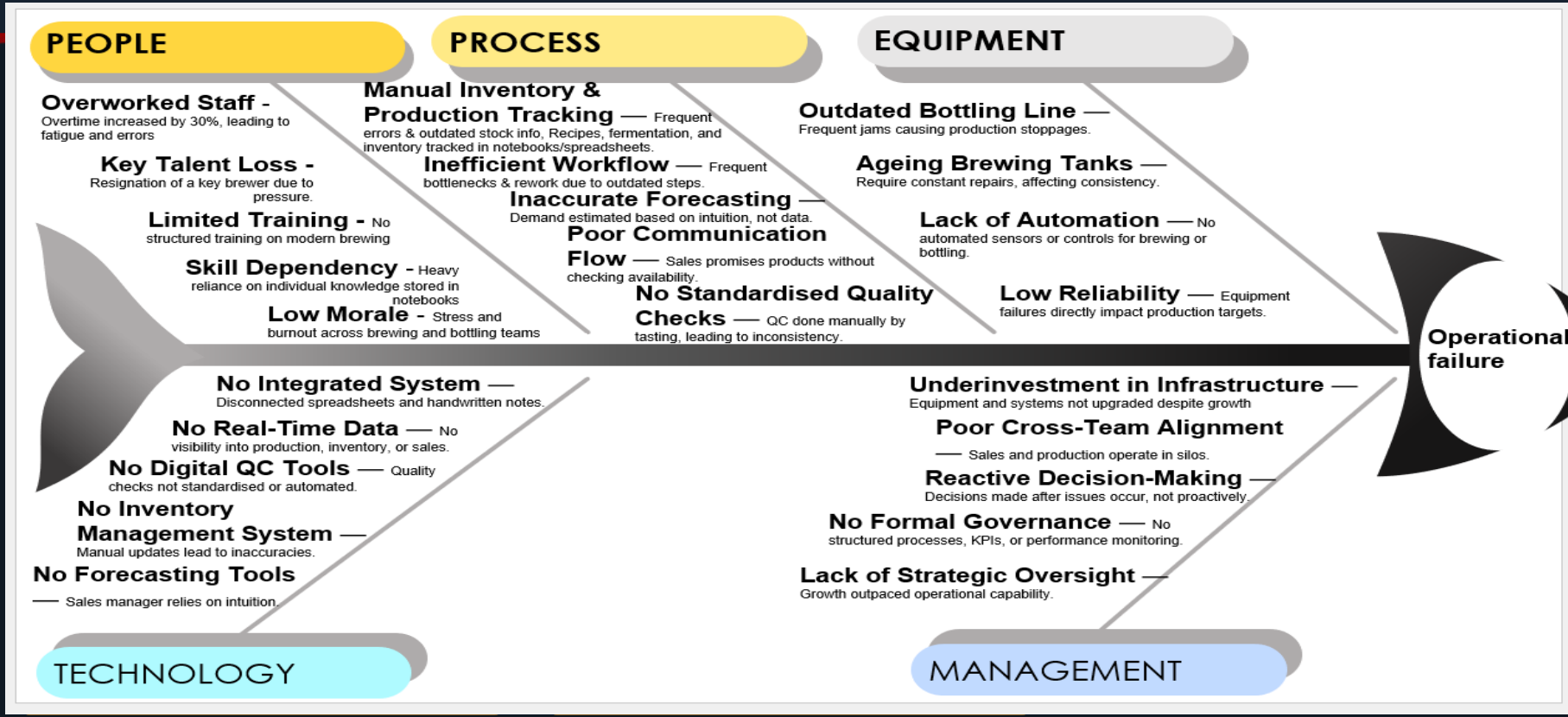
5 Data-Driven Decisions

No forecasting data, dashboards, or real-time inventory

6 Modern Infrastructure

Fragmented, outdated tech & equipment not fit for scale

FISHBONE DIAGRAM



Current State Process: Brewing to Bottling

PAIN POINTS

1
Demand Estimate
Manual intuition

2
Production Request
Phone call to brewer

3
Recipe Check
Handwritten notebooks

4
Brewing Begins
Fully manual

5
Fermentation Tracking
Notebook / spreadsheet

6
Quality Check
Manual taste test only

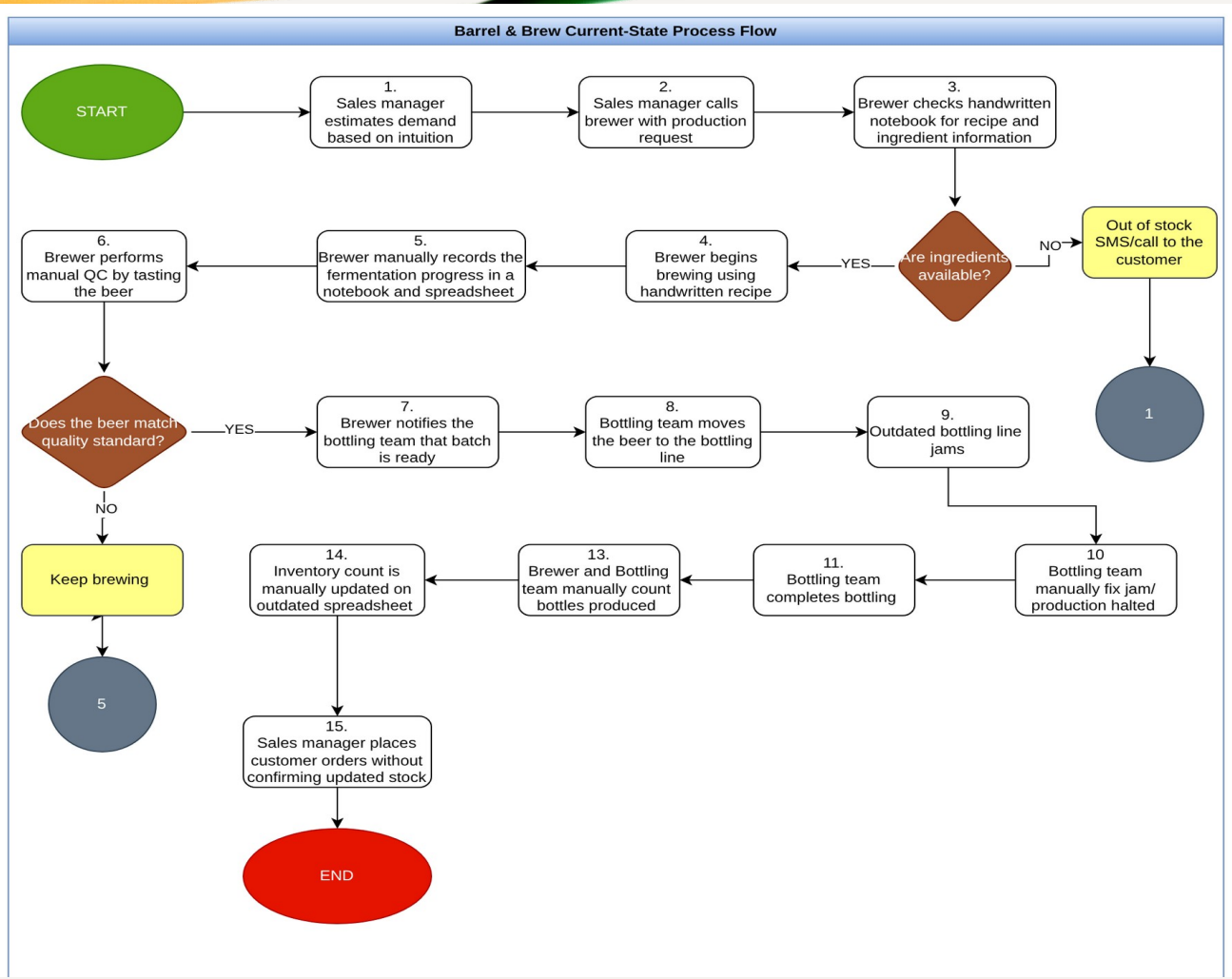
7
Notify Bottling
Ad hoc communication

8
Bottling Line
Jams → manual fix → delays

9
Stock Count
Manual bottle count

10
Order Placed
No stock check

= Pain point / breakdown risk


Barrel & Brew Current-State Process Flow
START
1.
Sales manager estimates demand based on intuition
2.
Sales manager calls brewer with production request
3.
Brewer checks handwritten notebook for recipe and ingredient information
Are ingredients available?
YES
NO
Out of stock SMS/call to the customer
1
4.
Brewer begins brewing using handwritten recipe
5.
Brewer manually records the fermentation progress in a notebook and spreadsheet
6.
Brewer performs manual QC by tasting the beer
Does the beer match quality standard?
YES
NO
7.
Brewer notifies the bottling team that batch is ready
8.
Bottling team moves the beer to the bottling line
9.
Outdated bottling line jams
10
Bottling team manually fix jam/ production halted
11.
Bottling team completes bottling
13.
Brewer and Bottling team manually count bottles produced
14.
Inventory count is manually updated on outdated spreadsheet
15.
Sales manager places customer orders without confirming updated stock
END
Keep brewing
5

Proposed Solution: Integrated Brewery Management System

Demand Forecasting

Automated weekly forecasts from distributor & restaurant data

Digital Recipe Mgmt

Versioned recipes with ingredient validation & sensor integration

Digital QC Workflow

Auto QC checkpoints, digital capture, hold/release capability

Automated Bottling

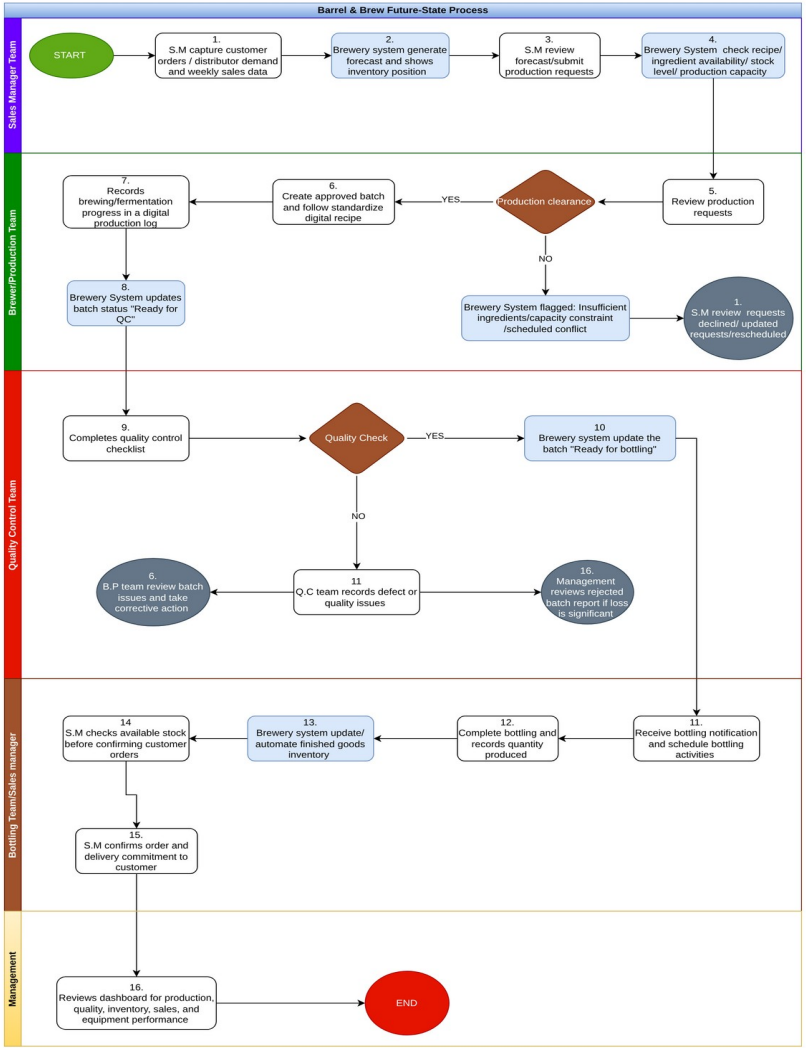
Jam detection, auto-stop, bottle count → live inventory sync

Real-Time Inventory

Auto-updates, low-stock alerts, batch & expiry tracking

Dashboards & Analytics

Production, sales, QC & inventory KPIs exportable in PDF / Excel


Barrel & Brew Future-State Process
Sales Manager Team
Brewer/Production Team
Quality Control Team
Bottling Team/Sales manager
Management
START
1. S.M capture customer orders / distributor demand and weekly sales data
2. Brewery system generate forecast and shows inventory position
3. S.M review forecast/submit production requests
4. Brewery System check recipe/ ingredient availability/ stock level/ production capacity
5. Review production requests
Production clearance
YES
NO
6. Create approved batch and follow standardize digital recipe
7. Records brewing/fermentation progress in a digital production log
8. Brewery System updates batch status "Ready for QC"
Brewery System flagged: Insufficient ingredients/capacity constraint /scheduled conflict
1. S.M review requests declined/ updated requests/rescheduled
9. Completes quality control checklist
Quality Check
YES
NO
10 Brewery system update the batch "Ready for bottling"
11 Q.C team records defect or quality issues
6. B.P team review batch issues and take corrective action
16. Management reviews rejected batch report if loss is significant
11. Receive bottling notification and schedule bottling activities
12. Complete bottling and records quantity produced
13. Brewery system update/ automate finished goods inventory
14 S.M checks available stock before confirming customer orders
15. S.M confirms order and delivery commitment to customer
16. Reviews dashboard for production, quality, inventory, sales, and equipment performance
END

Business Requirements Summary

FUNCTIONAL REQUIREMENTS

- Integrate distributor & restaurant sales data
- Auto-generate weekly demand forecasts
- Digital recipe storage with version control
- IoT sensor integration (temp, pH, gravity)
- Automated QC checkpoints with digital capture
- Bottling line: jam detection & auto-count
- Real-time inventory updates post-brewing/bottling
- Live dashboards — production, sales, QC, stock

NON-FUNCTIONAL REQUIREMENTS

Performance:	Real-time updates < 5 sec latency
Security:	GDPR compliant · Role-based access control
Scalability:	Supports 2× production growth in 3 years
Reliability:	System uptime ≥ 99.5%
Usability:	Intuitive for non-technical staff
Compatibility:	Mobile & tablet supported

Projected Benefits & Business Impact

50%

Reduction in production delays

100%

Accurate real-time inventory updates

30%

Reduction in overtime costs

OPERATIONAL

- Automated QC reduces batch failures
- Improved cross-team communication
- Real-time bottling & inventory sync

FINANCIAL

- Reduced waste & rework costs
- Lower overtime expenditure
- Optimised inventory reduces overstock

PEOPLE

- Reduced manual workload for brewers
- Improved staff morale & retention
- Clear workflows & accountability

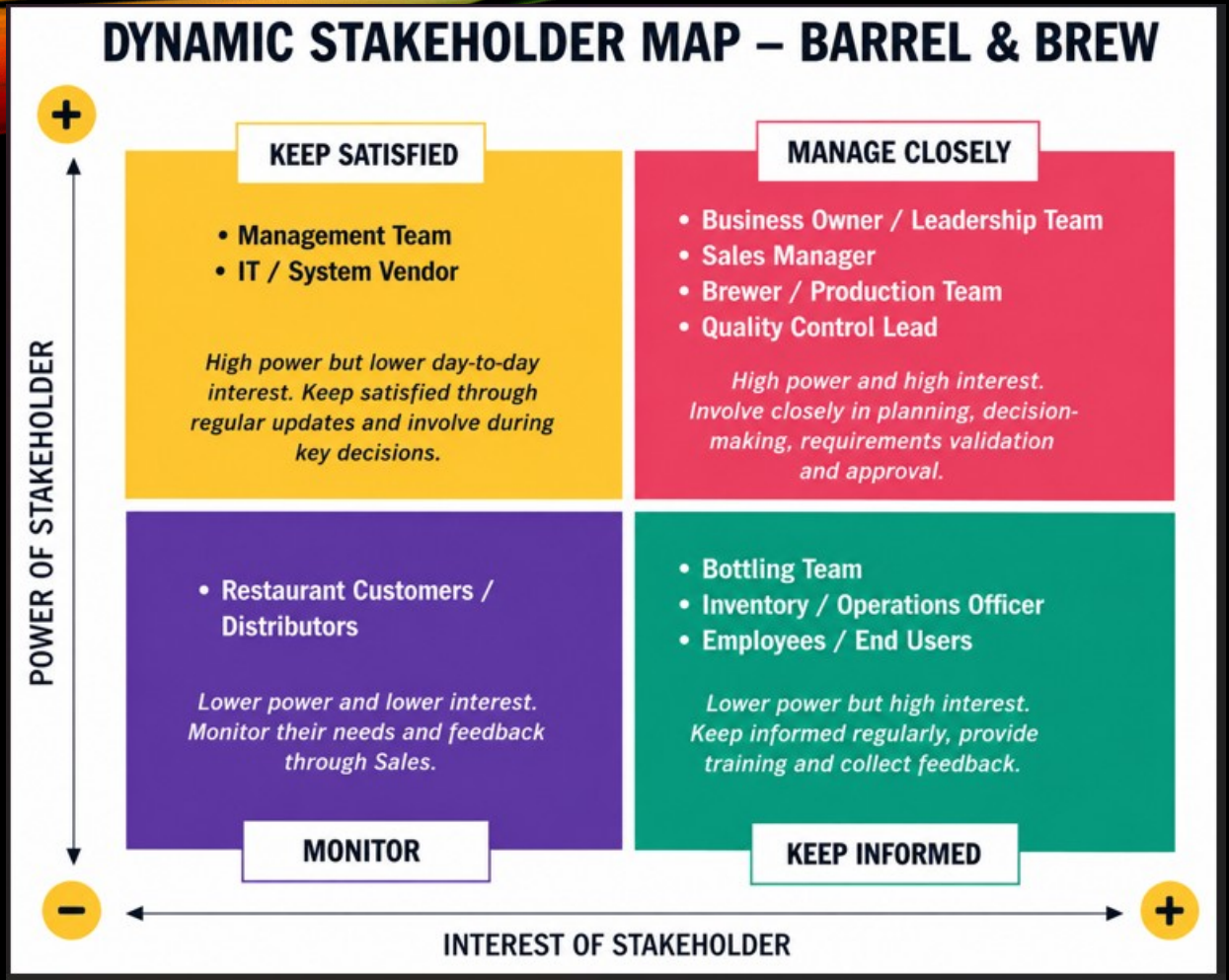
STRATEGIC

- Scalable, data-driven operating model
- Improved customer satisfaction
- Platform for sustainable growth

Stakeholder Register

Stakeholder	Role	Influence	Key Responsibility
CEO	Executive Sponsor	HIGH	Approve funding & strategic direction
Brewery Manager	Process Owner	HIGH	Validate requirements, oversee delivery
Sales Manager	Business Stakeholder	HIGH	Provide sales data, validate forecasting
IT Lead / Vendor	Technical Delivery	HIGH	Implement & support the solution
Quality Technician	QC Owner	MEDIUM	Define QC requirements & standards
Inventory Controller	Operations	MEDIUM	Manage inventory workflows
Brewers	End Users	MEDIUM	Use system daily; provide feedback
Bottling Team	End Users	MEDIUM	Operate new bottling equipment

DYNAMIC STAKEHOLDER MAP – BARREL & BREW



Next Steps

Proposed actions to bridge the gap

A	People:	Train staff on BMS tools, introduce SOPs, improve Sales–Production communication
B	Process:	Replace notebooks with digital recipe & QC management; automate inventory updates & forecasting
C	Equipment:	Replace outdated bottling line; upgrade brewing tanks with IoT sensors; automate bottle counting
D	Technology:	Implement integrated BMS; enable real-time dashboards; integrate distributor sales data
E	Management:	Establish KPIs, governance framework, and proactive planning cadence across all teams

Leadership approval required to proceed to vendor selection and solution design phase.

Thank You

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Questions & Discussion

Prepared by: Ikea Onyinyechi · Pod Polaris · Barrel & Brew Digital Transformation Project